UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 006 · Problems on Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. 1 · Testing homogeneity: the scaling test
2. 2 · Verifying Euler's theorem directly
3. 3 · When u itself is homogeneous: first and second order
4. 4 · When f(u) is homogeneous: the extension
5. Past-paper questions

1 Testing homogeneity: the scaling test

Definitions

Homogeneous function (scaling form). $u(x, y)$ is homogeneous of degree n if $u(\lambda x, \lambda y) = \lambda^n u(x, y)$. Here n can be positive, negative or even zero. S006·5 · T006b 00:00:52 · T006b 00:01:11

Not homogeneous. u is not homogeneous when $u(\lambda x, \lambda y) \neq \lambda^n u(x, y)$ for every n: no single power of λ can be taken out. S006·8 · T006b 00:04:59

Key points

1. Write $u(\lambda x, \lambda y)$, take the highest power of λ out of the numerator and the denominator, and cancel. ESA ★★ S006·5 · T006b 00:02:17
2. A λ inside a ratio cancels: $\sin^{-1}\frac{\lambda y}{\lambda x} = \sin^{-1}\frac{y}{x}$, so functions of y/x add degree 0. ESA ★★ S006·6 · T006b 00:03:34

3. $e^{\lambda x}$ cannot be written as $\lambda^n e^x$, so exponentials of x or y alone break homogeneity (slide 006·8). ESA ★★ S006·8 · T006b 00:04:40
4. Simplify first when you can: slide 006·5 is $\frac{(x^2+y^2)^2}{(x^2+y^2)^2} = 1$ and slide 006·7 is $\frac{xy(x+y)}{(x+y)^3} = \frac{xy}{(x+y)^2}$; both have degree 0. ESA ★ ADDED FOR CLARITY S006·5 · S006·7

Why?

Why does slide 006·8 fail the test?

$u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda(x-y)}$. The denominator gives λ^{-1}, but $e^{\lambda x} - e^{\lambda y}$ is not a power of λ times $e^x - e^y$. Figure G1.1 shows the ratio is not a straight line on log-log axes.

ADDED FOR CLARITY S006·8

2 Verifying Euler's theorem directly

Statements in the class material

> *"Euler's theorem: if $u = f(x, y)$ is a homogeneous function of degree n, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."*
>
> S006·10 · T006a 00:00:00

Key points

1. Verification is a check of the theorem, not a method: the lecture spends four minutes on slide 006·3; applying the theorem takes one line. ESA ★ T006a 00:04:57 · T006a 00:08:41
2. In slide 006·3, group the terms with the common denominator $(\sqrt{x} + \sqrt{y})^2$ and cancel one factor of $\sqrt{x} + \sqrt{y}$. ESA ★ T006a 00:02:59 · T006a 00:03:17

Exam traps

Exam trap. Slide 006·3 says "verify for $\cos^{-1}(\dots)$" but verifies for $Z = \cos u$. For u itself, $xu_x + yu_y = \frac{1}{2}\frac{\cos u}{-\sin u} = -\frac{1}{2}\cot u$, not $\frac{1}{2}u$. (ADDED FOR CLARITY) S006·3

3 When u itself is homogeneous: first and second order

Statements in the class material

> *"By Euler's theorem for a homogeneous function of degree n:* $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$ *."* S006·11

Key points

1. First prove that u is homogeneous and find n; then write both answers without differentiating. ESA ★★★ T006b 00:06:11 · T006b 00:06:45
2. Give the answer in terms of x and y too: $-u = -\frac{x-y}{x^2+y^2}$ and $2u = \frac{2(x-y)}{x^2+y^2}$. ESA ★★ S006·10 · S006·11
3. For a sum of parts with different degrees, apply the corollary part by part: $n_1(n_1-1)u_1 + n_2(n_2-1)u_2$. ESA ★★★ (added for clarity) ESA Jan-24 s3 1c · CW U1 P22

Exam traps

Exam trap. $n = -1$ gives $n(n-1) = (-1)(-2) = +2$, not -2: the second-order answer on slide 006·11 is $+2u$. (added for clarity) S006·11

4 When f(u) is homogeneous: the extension

Definitions

Extension function g(u). $g(u) = n\dfrac{f(u)}{f'(u)}$, where $f(u)$ is homogeneous of degree n in x and y.

S006·13 · S006·15 · S006·17

Statements in the class material

> *"For $f(u)$ a homogeneous function of degree n: $xu_x + yu_y = n\left(\dfrac{f(u)}{f'(u)}\right) = g(u)$, and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$."* S006·13 · S006·15 · S006·17

Key points

1. When u is $\sin^{-1}$, $\tan^{-1}$, $\log$ or e of something, apply the inverse function to both sides: that is $f(u)$. ESA ★★★ T006b 00:08:20 · T006b 00:09:55 · T006b 00:11:40
2. The degree belongs to $f(u)$, found by the scaling test on $f(u)$ (λ^4 on slide 006·12, λ on 006·14, $\lambda^{1/2}$ on 006·16). ESA ★★★ S006·12 · S006·14 · S006·16
3. "Whenever you have a complicated function, you can always use Euler's theorem if you feel that the function is homogeneous" — the lecture's closing advice for part I. ESA ★★ T006a 00:08:53
4. Three variables work the same way: on slide 006·4, $\sin u$ has degree -3, so $xu_x + yu_y + zu_z = -3\tan u$. ESA ★★ S006·4 · T006a 00:06:54

Why?

Why does the log case give -1 for the second order?
$f(u) = e^u$ gives $g = n\,e^u/e^u = n$, a constant. So $g' = 0$ and $g(g' - 1) = -n$; slide 006·14 has $n = 1$.

S006·15

Why can't Euler's theorem be applied to u directly on slides 006·12–17?
u is not homogeneous: for $u = \log\frac{x^2+y^2}{x+y}$, $u(\lambda x, \lambda y) = \log\lambda + u$, which is a shift, not a factor λ^n.
ADDED FOR CLARITY T006b 00:09:55

Exam traps

Exam trap. For $u = e^{(\cdots)}$ the function is $f(u) = \log u$, so $f' = 1/u$ and $f/f' = u\log u$: the answer is $4u\log u$, not $4\log u$. ADDED FOR CLARITY S006·13

Exam trap. In the second order use $g'(u)$, the derivative of g with respect to u, not $f'(u)$: for $g = \frac{1}{2}\tan u$, $g' = \frac{1}{2}\sec^2 u$. S006·17 · T006b 00:12:42

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler problems appear in six of them: Jul 2023 Q1.a (degree 5: $5u$, $20u$), May 2024 Q1.a and June 2024 Q1.b (the same $\tan^{-1}$ extension question), Jan 2024 set 3 Q1.c ($17\pi/2$), set 4 Q1.d ($9u$) and set 1 Q1.d (degree 0). Five are 6-mark questions. Testing homogeneity alone and verifying Euler's theorem by direct differentiation (topics 1–2) were not asked on their own; they appear only as the first step of a longer question.

3 · When u itself is homogeneous: first and second order

ESA Jul 2023 · 6 marks · UE22MA141A Q1.a · solved in W3.4

If $u(x,y) = x^3y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.c · solved in W3.7

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.d · solved in W3.6

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3} \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W3.8

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $xu_x + yu_y + zu_z$ (degree 0; also solvable by the chain rule).

4 · When f(u) is homogeneous: the extension

ESA May 2024 · 6 marks · UE22MA141A Q1.a · solved in W4.5

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 · 6 marks · UE23MA141A Q1.b · solved in W4.5

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).